

Chapter 2 supplementary material

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This document accompanies Chapter 2 of

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This supplement includes:

1. An illustration of the distinction between a partial effect (from a multiple regression model) and a marginal effect (from a simple regression model)
2. Brief descriptions of polynomial regression and variable transformations

Illustration of the distinction between a partial effect and a marginal effect

From the OLS formulas for the regression slope estimates (Equations 2.4 and 2.5), we can see computationally how the partial regression slope β_1 gives the relation between X_1 and Y while controlling X_2 . Researchers often expect that the effect of one regressor should be weaker when a second regressor is controlled. Indeed, that is the case for the *review* regressor in the movie example. Specifically, in Chapter 1, the estimated simple regression coefficient for *review* was 0.70, and then in Chapter 2, when the *length* variable was included in the model, the partial regression coefficient for *review* was lower, equaling 0.63.

It is entirely possible, however, for the partial effect of a given regressor, obtained by controlling one or more additional regressors, to be *stronger* (or even of the opposite sign) than the corresponding marginal effect for that regressor. One can determine that this scenario is possible by carefully considering different patterns of correlations among the regressors and outcome variable that would lead to a partial regression slope estimated with Equation 2.4 to be *larger* than the marginal slope for the same regressor estimated with Equation 1.6.

But perhaps the difference between a marginal and partial regression slope is most clearly illustrated using two-dimensional graphs in which the regressors to be controlled, X_2 , is dichotomous, as demonstrated below.

Partial effect weaker than marginal effect

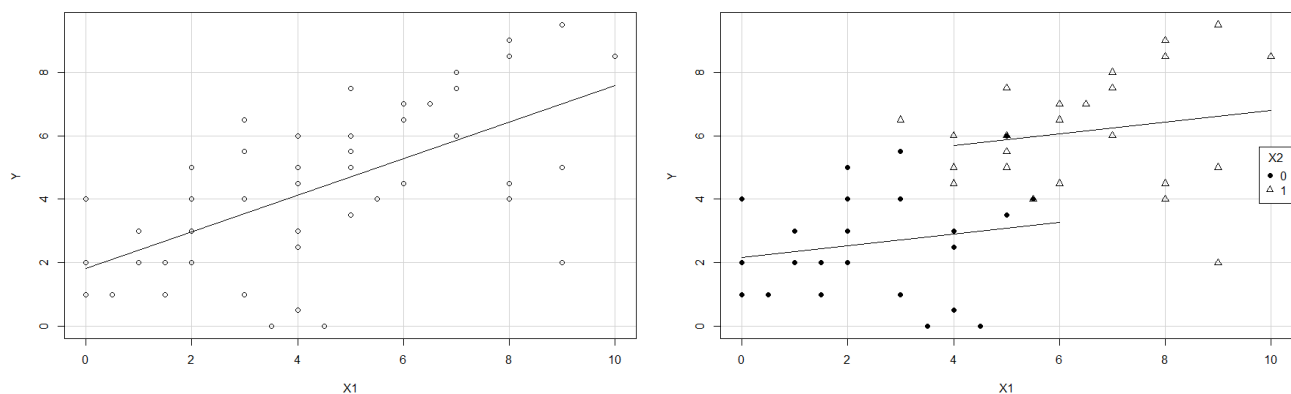


Figure S2.1

Scatterplots for contrived data with marginal regression slope for X_1 (left panel) and partial regression slope for X_1 , holding X_2 constant (right panel)

The left-hand panel of Figure S2.1 shows a scatterplot of made-up data in which the marginal relation between X_1 and Y is clearly strongly positive. For these data, the estimated marginal effect, that is, the slope of the simple regression of Y on X_1 , is $\hat{\beta} = 0.58$, which is significantly different from zero, $t(47) = 5.52, p < .0001$. But these results were obtained by ignoring a second variable, X_2 , which is dichotomous.¹ The right-hand panel of Figure S2.1 is a scatterplot of the same data, with different symbols used to distinguish the values of X_2 . Now we can see that *within* a given value of X_2 , the relation between X_1 and Y is not particularly strong. This plot illustrates what is meant by “holding X_2 constant:” one focuses on the relation between X_1 and Y within each value of X_2 . When we estimate the multiple regression model for these data, the estimated partial relation between Y and X_1 is $\hat{\beta}_1 = 0.18$, which is not significant, $t(46) = 1.30, p = .20$. Thus, for these data, the *partial* relation between X_1 and Y (i.e., the relation that *partials out* or *holds constant* X_2) is much weaker than the marginal relation between X_1 and Y .² The statistically significant marginal effect of X_1 becomes a non-significant partial effect when X_2 is controlled.

Partial effect stronger than marginal effect

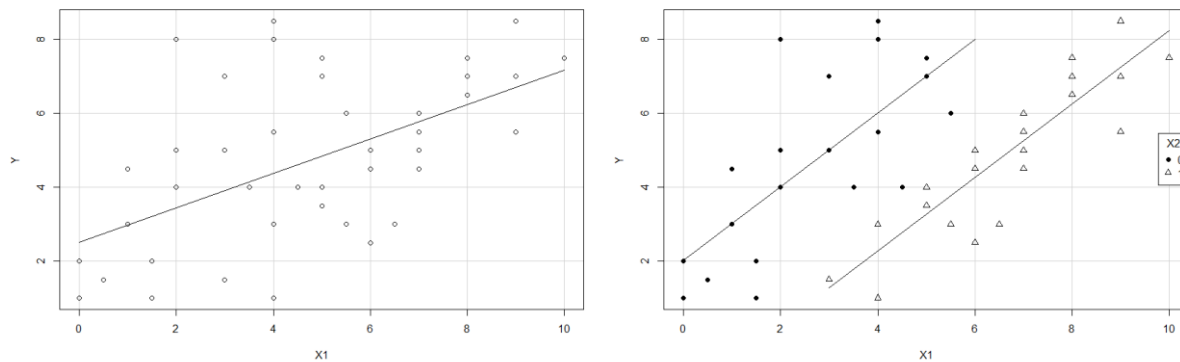


Figure S2.2

Scatterplots for contrived data with marginal regression slope for X_1 (left panel) and partial regression slope for X_1 , holding X_2 constant (right panel)

Figure S2.2 presents another made-up dataset. Like the data in Figure S2.1, the left-hand panel of Figure S2.2 clearly shows a positive marginal relation between X_1 and Y . Specifically, the slope of the simple regression of Y on X_1 is $\hat{\beta} = 0.46$, which is significant, $t(48) = 4.98, p < .0001$. But once again, these results were obtained by ignoring a second dichotomous regressor, X_2 . The right-hand panel of Figure S2.2 is a scatterplot of the same data, again with different symbols used to distinguish the values of X_2 . Now, it is apparent that the relation between X_1 and Y is even stronger within a given value of X_2 . When we estimate the two-predictor regression model for these data, the estimated partial relation between Y and X_1 is $\hat{\beta}_1 = 1.00$. Thus, for these data, the partial relation between X_1 and Y is much stronger than the marginal relation between X_1 and Y .³ These data provide an example of a phenomenon known as **enhancement**, which occurs when the partial effect of a given regressor is

¹ Recall from Chapter 1 that the relation between a dichotomous regressor and a continuous outcome is necessarily linear. Here, X_2 is dichotomous to simplify the demonstration, but the principles described readily generalize to a continuous version of X_2 .

² Note also that the data in Figure 2.3 do *not* depict an interaction between X_1 and X_2 . If there were an interaction, the relation between X_1 and Y would differ across the values of X_2 , but the partial regression slopes in the right-hand panel of Figure S2.1 are parallel, reflecting the lack of an interaction in the model. Interactions are addressed in Chapter 4.

³ Note once again that this difference also does *not* arise because of an interaction between X_1 and X_2 .

greater than the corresponding marginal effect.⁴ Although both the marginal and partial effects of X_1 are statistically significant in this example, it's also possible for a marginal effect to be nonsignificant while the corresponding partial effect controlling for one or more additional regressors is significant.

Positive marginal effect but negative partial effect

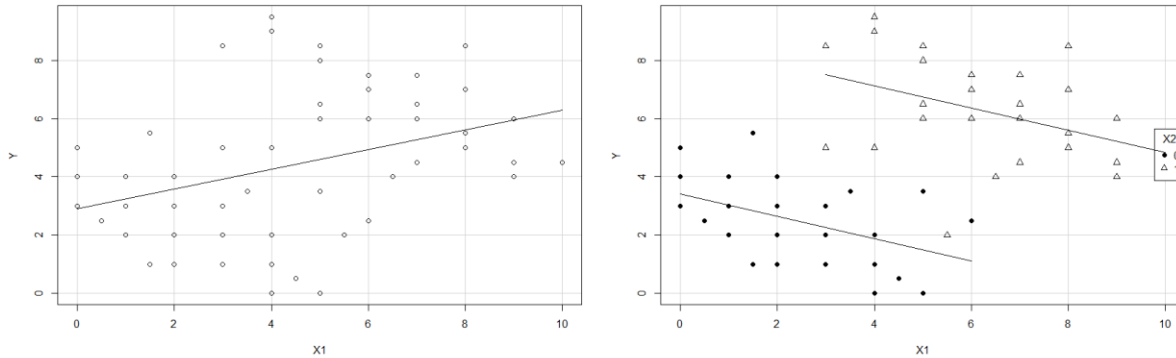


Figure S2.3

Scatterplots for contrived data with marginal regression slope for X_1 (left panel) and partial regression slope for X_1 , holding X_2 constant (right panel)

Finally, it is also possible for the partial relation between Y and X_1 to be negative when the marginal relation is positive (or vice versa), as the new set of made-up data in Figure S2.3 illustrates. Here again, the left-hand panel of the figure shows that the marginal relation between X_1 and Y is positive; here, $\hat{\beta} = 0.34$, $t(48) = 2.67$, $p = .01$. But now, as depicted in the right-hand side of Figure S2.3, when we hold X_2 constant, the partial relation between Y and X_1 is negative, in particular $\hat{\beta}_1 = -0.38$, which is also significant, $t(47) = -3.19$, $p = .003$.⁵ These data provide an example of a phenomenon known as *Simpson's paradox*, which occurs when the sign of a partial effect of a given variable is the opposite sign of the corresponding marginal effect.

Quadratic regression

Researchers commonly use **quadratic regression** model to represent a curvilinear relation between an outcome variable and regressor X_p . The quadratic regression equation for a single X is (S2.1) $Y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + \varepsilon_i$.

Note that this equation includes both a *linear term*, $\beta_1 X$, and a *quadratic term*, $\beta_2 X^2$. We can then use OLS to estimate the parameters β_0 , β_1 , and β_2 . Specifically, the term X^2 is treated as just another variable in the model; that is, this model is equivalent to

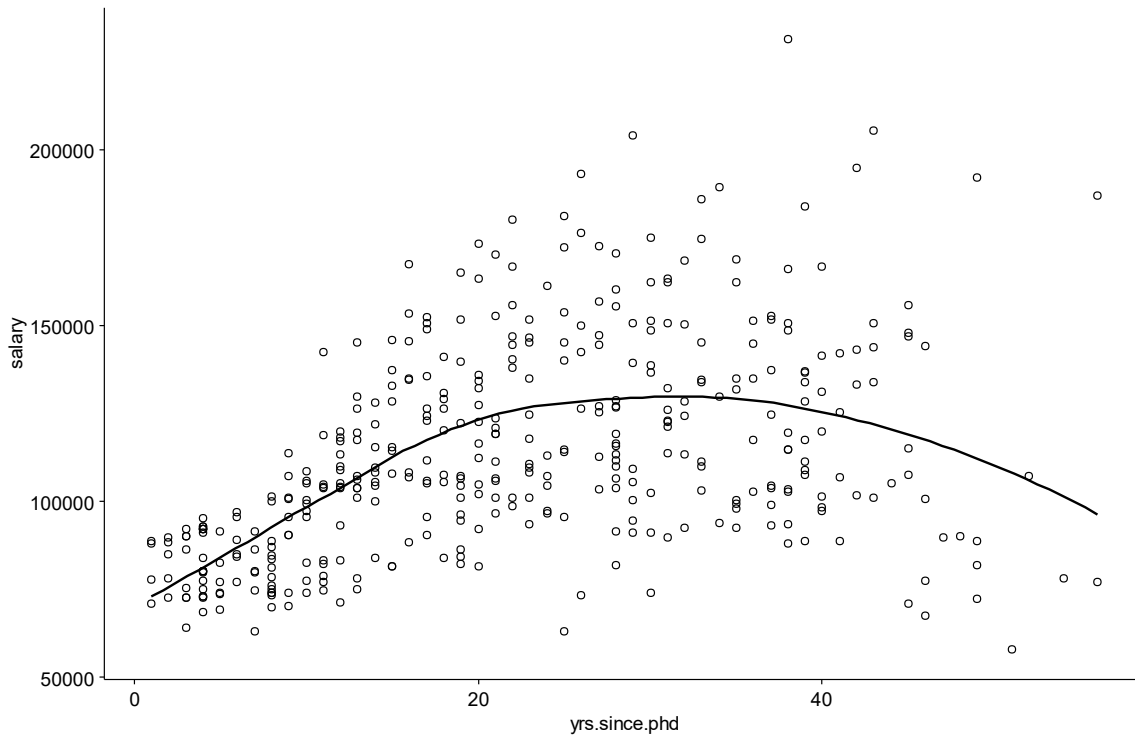
$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \varepsilon_i$$

with X_2 defined as X_1^2 . If $\hat{\beta}_2$ is large, then there is substantial curvature in the relation between X and Y such that a scatterplot of Y by X should depict a single bend in the data that is easily discerned visually. Of course, one can specify and estimate a model with nonlinear terms for one or more regressors while other regressors are modeled linearly. For example, we could simultaneously include both a linear relation between *review* and *profit* and a nonlinear, quadratic relation between *length* and *profit*.

⁴ Specifically, **enhancement** occurs when the standardized partial effect exceeds the standardized marginal effect while the total proportion of variance explained by the model (i.e., multiple R^2) is greater than $r_{YX_1} + r_{YX_2}$. **Suppression** is the situation in which the standardized partial effect exceeds the standardized marginal effect, but R^2 is less than $r_{YX_1} + r_{YX_2}$.

⁵ And once again, there is no interaction between X_1 and X_2 .

To illustrate quadratic regression, let's consider the `salaries` dataset that's available from the `carData` package in R (this dataset was also used in a supplemental analysis for Chapter 1). Viewing professor salary as the outcome variable and `yrs.since.phd` as the regressor, a scatterplot with a LOWESS curve shows a strong nonlinear association:



Next, we can estimate a quadratic model for salary with `yrs.since.phd` as the regressor, but before doing so, it will be helpful to **mean-center** the regressor to facilitate interpretation of the model's parameter estimates (centering is used extensively in Chapters 4 and 7):

$$X_{centered,i} = X_i - \bar{X}$$

That is, we create a new version of variable X by simply subtracting the sample mean of X from each observed value of X , and then use the new, centered variable in the regression model. Therefore, the quadratic model for salary is

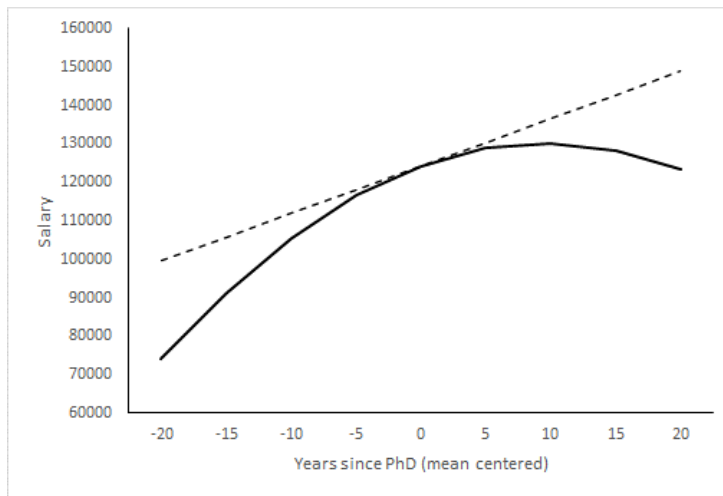
$$salary_i = \beta_0 + \beta_1(yrs.since.phd_{centered,i}) + \beta_2(yrs.since.phd_{centered,i})^2 + \varepsilon_i.$$

The estimated model (obtained with OLS) is

$$salary_i = 124,265.27 + 1231.24(yrs.since.phd_{centered,i}) - 63.74(yrs.since.phd_{centered,i})^2 + \varepsilon_i.$$

As with any regression model, the intercept is the predicted value of Y for a hypothetical case with a value of 0 on all regressors. Remember, however, that we mean-centered `yrs.since.phd` before estimating the model, and so a case with this centered regressor equal to 0 corresponds to a professor with the mean number of `yrs.since.phd`. Therefore, the intercept estimate of \$124,265.27 is the predicted salary for a professor with the mean number of `yrs.since.phd`.

Next, the *first-order coefficient* estimate (i.e., the linear term estimate), $\hat{\beta}_1 = 1,231.24$, is the slope of the line that is **tangent** to the quadratic function at $X = 0$:



Above, the solid curve is the fitted quadratic regression curve and the dashed line is tangent to the curve above $x=0$ (i.e., the line touches the curve at $X=0$ and nowhere else). The slope of the dashed line is $\hat{\beta}_1 = 1,231.24$. Because we centered *yrs.since.phd*, $\hat{\beta}_1 = 1,231.24$ is the *linear* regression effect that is specific to cases with the mean value of *yrs.since.phd*, which is approximately 22 years in this sample. Thus, among professors with around 22 years since PhD, each additional year predicts a difference of \$1,231.34 higher salary.

Now, how do we interpret the quadratic coefficient estimate, is $\hat{\beta}_2 = -63.74$? Roughly speaking, this estimate represents the extent of the nonlinear curvature (if $\beta_2 = 0$, there is no curvature; the model reduces to a straight line). Also, because the current $\hat{\beta}_2$ is negative, we know that the fitted quadratic model has an inverted-U shape (“concave down”); if $\hat{\beta}_2$ had been positive, then the fitted curve would have a U shape (“concave up”).

Specifically, though, $\hat{\beta}_2$ represents *half* of the amount that the tangent line slope changes when X increases by 1. Now, because $\hat{\beta}_1 = 1,231.24$ is the slope of the tangent line when $X=0$, then the slope of the tangent line when $X=1$ is

$$\hat{\beta}_1 + 2\hat{\beta}_2 = 1231.24 - 2(63.74) = 1103.76.$$

Why? For any expression of Y as function of X , the **first derivative** of the function gives the slope the tangent line at any particular value of X . For the quadratic regression function

$$\hat{Y}_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2,$$

the derivative is

$$\frac{dY}{dX} = \beta_1 + 2\beta_2 X.$$

Thus, when $X=0$, the tangent line slope is $\beta_1 + 2\beta_2 0 = \beta_1$.

Then, when $X=1$, the tangent line slope is $\beta_1 + 2\beta_2 1 = \beta_1 + 2\beta_2$.

More generally, $2\beta_2$ is the change in the tangent line slope corresponding to a 1-point increase in X .

Again, because we centered *years since PhD*, $\hat{\beta}_1 = 1,231.24$ is the linear regression effect that is specific to cases with the mean number of number of *yrs.since.phd*, which is approximately 22 years. Therefore, quadratic estimate $\hat{\beta}_2 = -63.74$ indicates that the linear effect corresponding to a hypothetical professor with 23 years since PhD is

$$\hat{\beta}_1 + 2\hat{\beta}_2 = 1231.24 - 2(63.74) = 1103.76.$$

Specifically, 1103.76 is the slope of a line that is tangent to the quadratic regression curve above $X=1$.

Returning to the movie example, Figure 2.3 in the main text shows that the residuals from the linear regression of *movie profit* on *review* and *length* have slight nonlinear relation with both regressors. Therefore, we might specify an alternative model that includes quadratic terms for both regressors:

$$profit_i = \beta_0 + \beta_1 review_i + \beta_2 review_i^2 + \beta_3 length_i + \beta_4 length_i^2 + \varepsilon_i$$

When this model is estimated with OLS, the resulting $R^2 = .13$ is significantly greater than $R^2 = .10$ obtained for the original model with only linear terms for *review* and *length*, $F(2, 604) = 11.17, p < .001$ (the original, linear model is nested within the quadratic model, allowing their comparison as a hierarchical regression). Regardless, the quadratic model results are consistent with the pattern suggested by the LOWESS curve in Figure 1.2: The curve bends upward only at higher levels of *review*, indicating that higher *review* scores tend to be associated with greater *profit* only among movies that have relatively high *review* scores in the first place.

Quadratic regression is an instantiation of the general class of **polynomial regression** models which can be used to represent a range of complex nonlinear associations. **Regression splines** represent a further generalization of polynomial regression and are commonly used in predictive modeling. In practice, however, these nonlinear models are extremely difficult to interpret and often represent an over-fitting to data such that models capture chance relations that are not replicable (regardless of statistical significance in a given sample); cross-validation methods are used with predictive modeling approaches to limit overfitting; see James et al. (2021) for a presentation of these techniques.

Transformations

The residuals from a regression model are often (but not always) non-normal when the outcome variable is itself non-normal. It is common to *transform* a non-normal outcome variable to create a new variable with a distribution more closely approximating a normal distribution; the transformed variable is then modeled instead of the original, raw-scale outcome. Modeling a transformed outcome variable might then produce a residual distribution that more closely approximates a normal distribution and can also improve heterogeneous variance and reduce the influence of outlying cases.

Commonly, a variable with strong positive skewness is transformed using a log transformation or a square-root transformation. That is, if Y_1 is positively skewed, one creates a new variable, $Y_2 = \ln(Y_1)$ or $Y_2 = \sqrt{Y_1}$, and then Y_2 is used as the outcome variable for a linear regression analysis. Formal routines such as the *Box-Cox* procedure can determine the optimal transformation for a given non-normally distributed variable (see Fox, 2016, pp. 55-75). Regardless of the transformation chosen, it is critical that the rank-ordering of Y not be affected. Furthermore, adding a constant as a part of a transformation can help avoid illegal arithmetical operations, such as taking a logarithm of 0 or the square root of a negative number.

Researchers are often hesitant to analyze transformed variables because of added interpretational complexity. For example, if a log-transformed Y is regressed on X , then the slope estimate $\hat{\beta}$ is the predicted difference in the *log* of Y associated with a one-unit difference in X . Yet, the effect of X can be expressed in terms of the original scale of Y by calculating the antilog, or exponentiation, of $\hat{\beta}$; that is, $\exp(\hat{\beta}) = e^{\hat{\beta}}$. Then, $\exp(\hat{\beta})$ represents the predicted difference in the original scale of Y per one-unit higher X . But, if a researcher finds that major substantive conclusions are very similar regardless of whether a variable is transformed, then it may be preferable to report results from the model of the original, untransformed variable.